

Practical Project Report: OSPF Configuration and Deployment with Gateway of Last Resort

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Date of Execution:	2026-06-28	Simulation Environment:	Cisco Packet Tracer
Lab Title:	OSPF Routing with Gateway of Last Resort (ASBR Advertisement)		

1. Project Description and Topology Overview

The objective of this practical lab is to construct and secure an integrated dynamic routing environment utilizing **OSPF Area 0** across four core internal routers (R1, R2, R3, and R4). Additionally, it simulates external internet connectivity by linking the main edge router, **R1**, to an Internet Service Provider router (**ISPR1**) via a public IP subnet range.

In this scenario, the role of an **Autonomous System Boundary Router (ASBR)** is implemented on R1. A static default route pointing toward the ISP gateway next-hop is established, which is subsequently injected and dynamically propagated to all downstream routers within the internal OSPF domain. This setup guarantees that all local area network nodes automatically acquire a path to external networks without relying on manual configuration.

IP Addressing Plan

Router	Interface	IP Address	Subnet Mask	Connected Link / Target Network
R1 (ASBR)	GigabitEthernet3/0	203.0.113.1	255.255.255.252 (/30)	External Internet Link (Link to ISPR1)
	GigabitEthernet0/0	10.0.12.1	255.255.255.252 (/30)	Internal Transit Link to R2
	FastEthernet1/0	10.0.13.1		Internal Transit Link to R3

Router	Interface	IP Address	Subnet Mask	Connected Link / Target Network
			255.255.255.252 (/30)	
R2	GigabitEthernet0/0	10.0.12.2	255.255.255.252 (/30)	Internal Transit Link to R1
	FastEthernet1/0	10.0.24.1	255.255.255.252 (/30)	Internal Transit Link to R4
R3	FastEthernet1/0	10.0.13.2	255.255.255.252 (/30)	Internal Transit Link to R1
	FastEthernet2/0	10.0.34.1	255.255.255.252 (/30)	Internal Transit Link to R4
R4	FastEthernet1/0	10.0.24.2	255.255.255.252 (/30)	Internal Transit Link to R2
	FastEthernet2/0	10.0.34.2	255.255.255.252 (/30)	Internal Transit Link to R3
	GigabitEthernet0/0	192.168.4.254	255.255.255.0 (/24)	Local Area Network (LAN Gateway)

2. Configuration Commands Applied to Edge Router R1

The interface facing the ISP is configured, an active static default route is established to avoid recursive lookup issues, and the OSPF process is initialized to inject and distribute the default path to the rest of the enterprise topology:

```
R1(config)# interface GigabitEthernet3/0
R1(config-if)# ip address 203.0.113.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 1.1.1.1 0.0.0.0 area 0
R1(config-router)# network 10.0.12.0 0.0.0.3 area 0
R1(config-router)# network 10.0.13.0 0.0.0.3 area 0
R1(config-router)# passive-interface Loopback0
R1(config-router)# default-information originate
R1(config-router)# exit

R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

3. Verification and Factual Output Analysis

A. Edge Router R1 Verification Outputs

Executing the command **show ip protocols** confirms that the edge node recognizes its operational border role within the autonomous domain, generating external LSA records:

It is an autonomous system boundary router

```
R1(config)# do sh ip prot
```

```
Routing Protocol is "ospf 1"
```

```
Outgoing update filter list for all interfaces is not set
```

```
Incoming update filter list for all interfaces is not set
```

```
Router ID 1.1.1.1
```

```
It is an autonomous system boundary router
```

```
Redistributing External Routes from,
```

```
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

```
Maximum path: 4
```

```
Routing for Networks:
```

```
1.1.1.1 0.0.0.0 area 0
```

```
10.0.12.0 0.0.0.3 area 0
```

```
10.0.13.0 0.0.0.3 area 0
```

```
Passive Interface(s):
```

```
Loopback0
```

```
Routing Information Sources:
```

Gateway	Distance	Last Update
1.1.1.1	110	00:12:24
2.2.2.2	110	00:12:24
3.3.3.3	110	00:14:59
4.4.4.4	110	00:12:43

```
Distance: (default is 110)
```

Reviewing the internal routing structure via **show ip route** validates the complete installation of the default route entry **S***, mapping the Gateway of last resort successfully:

```

R1(config)# do sh ip rou
...
Gateway of last resort is 203.0.113.2 to network 0.0.0.0

    1.0.0.0/32 is subnetted, 1 subnets
C      1.1.1.1 is directly connected, Loopback0
    3.0.0.0/32 is subnetted, 1 subnets
O      3.3.3.3 [110/2] via 10.0.13.2, 00:19:21, FastEthernet1/0
    4.0.0.0/32 is subnetted, 1 subnets
O      4.4.4.4 [110/3] via 10.0.12.2, 00:13:24, GigabitEthernet0/0
        [110/3] via 10.0.13.2, 00:13:24, FastEthernet1/0
    10.0.0.0/30 is subnetted, 4 subnets
C      10.0.12.0 is directly connected, GigabitEthernet0/0
C      10.0.13.0 is directly connected, FastEthernet1/0
O      10.0.24.0 [110/2] via 10.0.12.2, 00:13:24, GigabitEthernet0/0
O      10.0.34.0 [110/2] via 10.0.13.2, 00:15:56, FastEthernet1/0
    203.0.113.0/30 is subnetted, 1 subnets
C      203.0.113.0 is directly connected, GigabitEthernet3/0
S* 0.0.0.0/0 [1/0] via 203.0.113.2

```

B. Internal Core Router R3 Output Analysis

As a consequence of the dynamic OSPF distribution mechanism, the database sync translates into a populated routing table on R3, appending an **O*E2** entry representing an external Type 2 metric path:

```
R3# sh ip rou
...
Gateway of last resort is 10.0.13.1 to network 0.0.0.0

    1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/2] via 10.0.13.1, 00:25:48, FastEthernet1/0
    3.0.0.0/32 is subnetted, 1 subnets
C       3.3.3.3 is directly connected, Loopback0
    4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/2] via 10.0.34.2, 00:22:06, FastEthernet2/0
    10.0.0.0/30 is subnetted, 4 subnets
O       10.0.12.0 [110/2] via 10.0.13.1, 00:19:46, FastEthernet1/0
C       10.0.13.0 is directly connected, FastEthernet1/0
O       10.0.24.0 [110/2] via 10.0.34.2, 00:20:06, FastEthernet2/0
C       10.0.34.0 is directly connected, FastEthernet2/0
O*E2 0.0.0.0/0 [110/1] via 10.0.13.1, 00:07:34, FastEthernet1/0
```

C. LAN Distribution Router R4 Output Analysis

Router R4 sits at the base of the topology, holding equal-cost paths (ECMP) toward the internet gateway via both transit layers (R2 and R3). It balances outbound traffic symmetrically over both paths:

```

R4# sh ip rou
...
Gateway of last resort is 10.0.24.1 to network 0.0.0.0

    1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/3] via 10.0.24.1, 00:21:56, FastEthernet1/0
        [110/3] via 10.0.34.1, 00:21:56, FastEthernet2/0
    3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/2] via 10.0.34.1, 00:24:31, FastEthernet2/0
    4.0.0.0/32 is subnetted, 1 subnets
C       4.4.4.4 is directly connected, Loopback0
    10.0.0.0/30 is subnetted, 4 subnets
O       10.0.12.0 [110/2] via 10.0.24.1, 00:21:56, FastEthernet1/0
O       10.0.13.0 [110/2] via 10.0.34.1, 00:24:31, FastEthernet2/0
C       10.0.24.0 is directly connected, FastEthernet1/0
C       10.0.34.0 is directly connected, FastEthernet2/0
C       192.168.4.0/24 is directly connected, GigabitEthernet0/0
O*E2 0.0.0.0/0 [110/1] via 10.0.24.1, 00:09:46, FastEthernet1/0
        [110/1] via 10.0.34.1, 00:09:46, FastEthernet2/0

```

D. Transit Core Router R2 Output Verification

```
R2# sh ip rout
...
Gateway of last resort is 10.0.12.1 to network 0.0.0.0

    1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/2] via 10.0.12.1, 00:22:56, GigabitEthernet0/0
    2.0.0.0/32 is subnetted, 1 subnets
C       2.2.2.2 is directly connected, Loopback0
    3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/3] via 10.0.24.2, 00:22:56, FastEthernet1/0
        [110/3] via 10.0.12.1, 00:22:56, GigabitEthernet0/0
    4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/2] via 10.0.24.2, 00:23:18, FastEthernet1/0
    10.0.0.0/30 is subnetted, 4 subnets
C       10.0.12.0 is directly connected, GigabitEthernet0/0
O       10.0.13.0 [110/2] via 10.0.12.1, 00:22:56, GigabitEthernet0/0
C       10.0.24.0 is directly connected, FastEthernet1/0
O       10.0.34.0 [110/2] via 10.0.24.2, 00:23:18, FastEthernet1/0
O*E2 0.0.0.0/0 [110/1] via 10.0.12.1, 00:10:46, GigabitEthernet0/0
```

4. Engineering Conclusions & Lessons Learned

The infrastructure converges flawlessly following the deployment of appropriate interface matching parameters. Primary engineering insights derived from this implementation include:

- **Resolution of Recursive Lookup Faults:** A static default route next-hop must reside within an explicitly reachable, directly connected subnet structure. Failing this, the routing engine drops the path candidacy, halting propagation entirely.
- **Scalable Dynamic injection:** Utilizing default-information originate abstracts administrative overhead, preventing manual misconfigurations across scaling core frameworks.
- **Symmetric Routing Redundancy:** The lower boundaries showcase proper execution of Equal-Cost Multi-Pathing (ECMP), providing optimized load redistribution and deterministic fault-recovery windows.